reveals how actions influence task outcomes, the agent's environment, and its internal states, contributing to efficient decision making. Lastly, grasping action generation methods equips us with the knowledge to create actions manually, recall them from memory, or follow predefined plans, enhancing our ability to effectively shape agent behavior and learning processes.

Figure 1.6 shows the component knowledge and memory in more detail. Agents use knowledge and memory to annotate context with the most pertinent information while limiting the number of tokens used. Knowledge and memory structures can be unified, where both subsets follow a single structure or hybrid structure involving a mix of different retrieval forms. Knowledge and memory formats can vary widely from language (e.g., PDF documents) to databases (relational, object, or document) and embeddings, simplifying semantic similarity search through vector representations or even simple lists serving as agent memories.

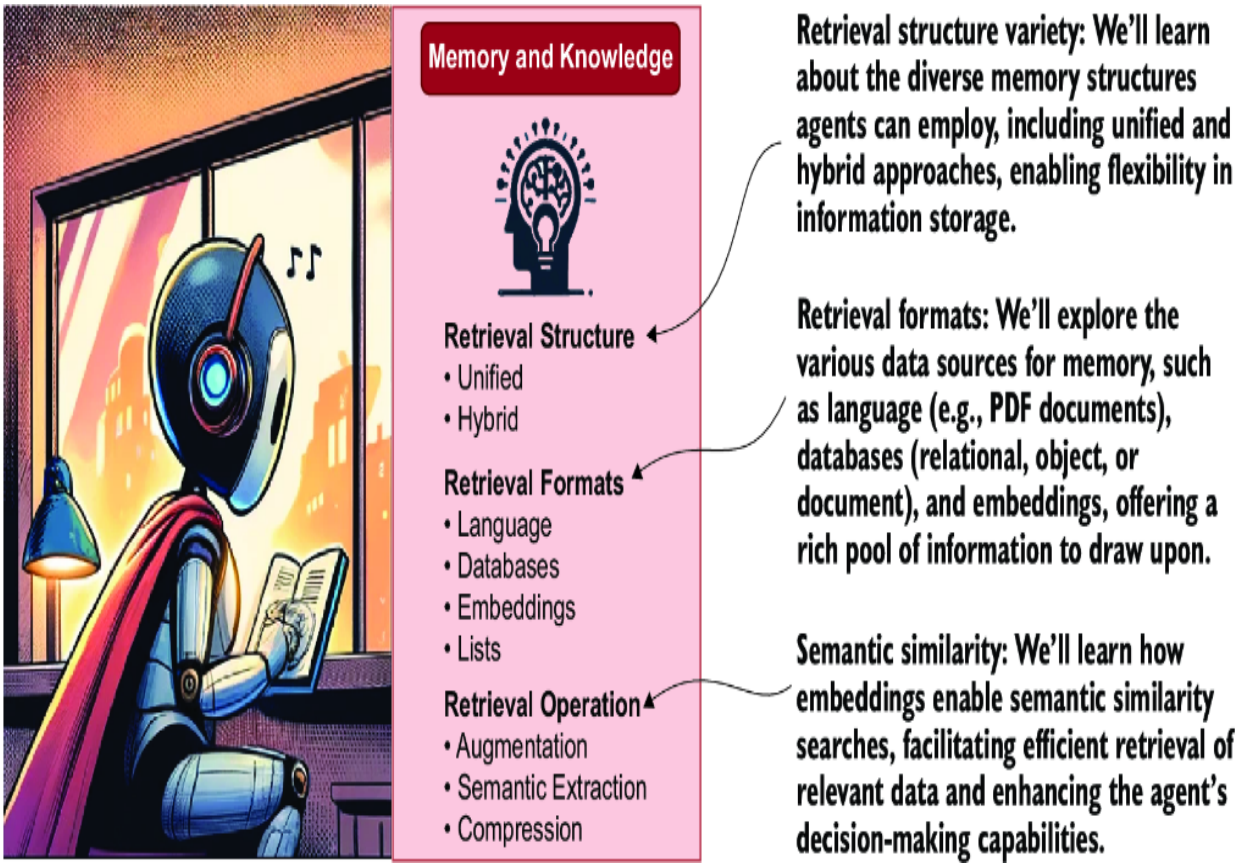


Figure 1.6 Exploring the role and use of agent memory and knowledge